

# Bioinformatics\_Bootcamp\_Assessment\_01

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## Question 1:

When a researcher finds a DNA sequence that might be linked to a genetic disorder the first thing to do is to see if this sequence has been reported before. To do this I would suggest using the NCBI GenBank, RefSeq and ClinVar databases.

The NCBI GenBank is a public collection of nucleotide sequences. It has DNA sequences sent in by researchers from over the world. By using the BLAST tool to search for the sequence in GenBank the researcher can find identical sequences that have been reported before. This helps figure out if the sequence's new or already known.

The RefSeq database provides reviewed reference sequences for genes, transcripts and proteins. Unlike GenBank, which accepts submissions RefSeq has records that have been checked making it a trustworthy source for finding well-characterized genes and variants.

The ClinVar database is especially important because it focuses on the connection between variants and human diseases. If the sequence has a mutation linked to a disorder ClinVar might have information about its clinical significance whether it causes disease and if it has been reported before.

For example if the sequence has a mutation in the CFTR gene that is linked to fibrosis ClinVar might say if this variant is known to cause disease. Together these databases help researchers find out if the sequence has been reported before and if it has any known importance.

## Question 2:

To compare the genome of a disease-causing bacterium with those of species I would suggest using the NCBI Genome, Ensembl Bacteria and the KEGG Database.

The NCBI Genome Database has draft genome sequences from thousands of bacterial species. It lets researchers get genome data and compare gene content between organisms. This is useful for finding genes that're the same across related bacterial species. The Ensembl Bacteria database provides genome annotations and tools for comparing genomes. Researchers can look at gene locations how they evolved and genes that are similar among bacterial species. Genes that are the same across species are often important for cellular functions. The KEGG database has information about



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metabolic pathways. What genes do. After finding genes that're the same researchers can use KEGG to find out which biological pathways those genes are part of.

For example if several related bacterial species have a gene that's the same and is involved in cell wall synthesis this gene might be crucial for the bacterium to survive and could be a target for drugs. By combining information from NCBI Genome, Ensembl Bacteria and KEGG scientists can find genes that're the same and understand what they do.

#### Question 3:

When a biotechnology company finds a protein from a newly discovered microorganism and wants to know if similar proteins exist in other species I would suggest using UniProt, NCBI Protein and Pfam.

The UniProt database is one of the complete protein databases available. It has protein sequences, descriptions of what they do information about domains and links to papers. Searching for the protein sequence in UniProt can reveal proteins found in other organisms. The NCBI Protein database has protein sequences from species. Using BLASTP to search this database can help find proteins with amino acid sequences and estimate how they are related. The Pfam database focuses on protein families and conserved domains. Even if the exact protein has not been reported before Pfam can find domains within the sequence. These domains often give clues about the proteins role.

For example if the protein has a domain that binds ATP and is commonly found in transport proteins researchers can guess that it might be involved in transport. Similar proteins found in bacteria, fungi or plants might support this idea. Together these databases help find out if the protein exists in species how it has evolved and what it might do.

#### Question 4:

When a scientist has a protein sequence and wants to know what it does I would suggest using UniProt, Pfam, InterPro and the Protein Data Bank (PDB).

The UniProt database provides notes about protein function, biological processes, where it is in the cell and related papers. Searching for sequences can find proteins with known functions that are similar to the unknown sequence. The Pfam database finds conserved protein domains and families. Since domains are often linked to biological functions finding a known domain can provide strong evidence about the proteins role.



The InterPro database combines information from protein databases, including Pfam and SMART. It predicts domains, motifs and protein families giving a broad view of the proteins characteristics. The PDB has experimentally determined three- protein structures. If the proteins structure is similar to a known protein it might give clues about its function even if the sequence is not very similar.

For example if the sequence has a kinase domain found by Pfam and InterPro and its structure is similar to a known kinase in PDB the protein is likely involved in adding groups and cell signaling. Using these databases together provides a way to predict protein function.

Question 5:

When given an unknown biological sequence the first task is to determine if it is DNA, RNA or protein. I would follow an approach using several biological databases and tools.

Look at the sequence composition. If the sequence only has A, T, C and G it is probably DNA. If it has A, U, C and G it is likely RNA. If it has amino acid symbols like M, K, L and F it is probably a protein sequence.

Do a BLAST search. For DNA or RNA sequences I would use BLASTN against the NCBI GenBank database. For protein sequences I would use BLASTP against the NCBI Protein database. This helps find sequences and possible biological origins.

If the sequence seems to be DNA or RNA I would look at GenBank and RefSeq for notes. If it is a protein sequence I would search UniProt for information and similar proteins.

For protein sequences I would use Pfam and InterPro to find conserved domains and predict function.

If necessary I would consult PDB for information and KEGG for pathway analysis.

For example if BLAST finds a strong match to an enzyme, UniProt and Pfam can reveal its function while KEGG can show the metabolic pathway it is part of. This step-, by-step approach allows for identification and characterization of an unknown biological sequence.



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